Marco Maida

Senior Machine Learning Engineer

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I'm a software and machine learning engineer with over **ten years of experience** delivering systems across **industrial software**, **videogames**, **real-time systems**, **formal verification**, and **autonomous vehicles**.

At **Wayve**, I develop real-time inference systems for autonomous vehicles. My work covers model and runtime co-design, heterogeneous accelerators, numerical validation, and latency optimisation. I work primarily in **C++**, **Rust**, and **Python**, with an emphasis on measurable performance, correctness, and production constraints.

Professional Experience

Wayve Ltd

London, United Kingdom

- **Senior Machine Learning Engineer** May 2026 → Present
  Co-design models and runtime architectures for real-time execution on resource-constrained automotive accelerators. Partition workloads across CPUs, GPUs, and NPUs; start model components as their sensor inputs become available; and maintain validation suites for numerical precision. Optimise end-to-end latency, throughput, and resource utilisation. *(C++, Python)*.
- **Tech Lead, Inference Performance** Dec 2024 → May 2026
  Led a team of four responsible for Wayve's on-board inference stack. Extended the platform across multiple inference frameworks and accelerators while supporting the same execution path in every vehicle and simulation environment. Owned latency, reliability, and technical delivery for the core inference component. *(C++, Rust, Python)*.
- **Senior Software Engineer** Dec 2022 → Dec 2024
  Joined a seven-person team building Wayve's second-generation on-board platform and shipping it to its first two vehicles. Built the initial real-time inference pipeline; Lidar, GNSS, and microphone stacks; data collection and upload features; and the vehicles' internal software network. *(ROS2, Linux, C++, Rust, Python)*.

34BigThings

Turin, Italy

Software Engineer – Videogames Aug 2016 → Sep 2019

Shipped five single-player and online multiplayer titles across Steam, PS4, Xbox One, Nintendo Switch, and mobile. Developed game infrastructure, AI, gameplay systems, and production tools in Unity and Unreal Engine. *(Unity3D, C#, Unreal Engine, C++)*.

Maserati / Choralia

Turin, Italy

Software Engineer (Contractor) – Games and Simulation Sep 2016 → May 2017

Delivered two B2B projects: a browser and mobile educational game, and a 3D product-visualisation tool. Led implementation and supervised an external artist. *(Unity3D, C#, JavaScript)*.

R.O. srl

Software Engineer – Industrial Software Jun 2013 → Aug 2016

Developed software for glass-processing factories covering order tracking, machine scheduling, waste reduction, and logistics. Managed four junior engineers during the third year. *(C, C++, C#, SQL)*.

Education and Research

Research Internship Jun 2022 → Aug 2022
Bloomberg LP
I worked on accelerating SAT solving using GPUs. *(C++, CUDA)*.

PhD Student Sep 2019 → Oct 2022
Max Planck Institute
My research focused on formal verification and real-time systems, specifically verifying the timeliness of software systems. I published three papers and mentored three interns. *(COQ, C, Rust)*.

Master's in Computer Science Nov 2019 → Jul 2022
Technische Universität Kaiserslautern
I specialized in real-time systems as part of a joint master's and PhD program.

Bachelor's in Computer Science Sep 2016 → Jun 2019
Università degli Studi di Torino
I specialized in computability and formal methods.

Selected Publications

Claycode: Stylable and Deformable 2D Scannable Codes 2025
SIGGRAPH 2025
★*Journal publication in Transactions of Graphics. Selected for CAF trailer.*
We introduced Claycodes, a new type of visual code that breaks free from traditional rigid grids like QR codes. Claycodes encode data as a tree structure and allow full customizability and deformation, with real-time scanning capabilities.
(https://arxiv.org/abs/2505.08666).

From Intuition to Coq: A Case Study in Verified Response-Time Analysis, FIFO Scheduling 2022
RTSS 2022
We developed a formally verified response-time analysis for FIFO schedulers, challenging traditional pen-and-paper methods.
(https://people.mpi-sws.org/~kbedarka/rtss22.pdf).

Foundational Response-Time Analysis as Explainable Evidence of Timeliness 2021
ECRTS 2022
★*Outstanding Paper Award.*
I developed POET, a tool for formally verified worst-case scenario timing analysis.
(https://drops.dagstuhl.de/opus/volltexte/2022/16336/pdf/LIPIcs-ECRTS-2022-19.pdf).

Selected Open-Source Projects

Claycode 2025
https://claycode.io
Inventor and lead contributor. Claycode includes a generator website, an Android scanner app, and a large amount of research code.

PROSA 2020 → 2022
https://gitlab.mpi-sws.org/RT-PROOFS/rt-proofs
Main contributor to PROSA, one of the leading formally-verified frameworks in the real-time systems community.

Treecode 2022
www.maida.me/treecode
The precursor to Claycode, a novel 2D scannable code that encodes messages as unique trees.

POET 2021
https://gitlab.mpi-sws.org/RT-PROOFS/POET
First-ever implementation of a foundational response-time analysis tool, created as part of my first academic publication.

Fast Mobile Cycle (FMC) Framework and Toolkit 2018
www.github.com/34OpenThings
Developed a Unity3D framework that accelerates production-ready casual game creation, paired with a Python toolkit for bulk operations.